

Source-Locked Integration at the Frontier Occurrence Lift: Four-Map Completion, an Actual Möbius Core, and Split Endpoint Ledgers

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July 2026

Abstract

TPC-143–150 pursue the two openings left by the MVP4 audit. The structural branch asks what must be added after the complete cut archive in order to define determinant and zero-mode quotients, physical grouping, and the prescribed-shift selector. The arithmetic branch replaces a Boolean squarefree expansion by an exact quotient lift and reaches a genuine fixed-two Möbius periodic core in a small-polylogarithmic almost-scale corridor.

We integrate both branches without escalating either result. The common structural object is a conservative row-separated matrix

$$L_X : \mathbb{C}^{\mathcal{C}_X^{\text{ns}}} \longrightarrow \mathbb{C}^{\mathcal{O}_X}, \quad \mathbf{1}^T L_X = \mathbf{1}^T,$$

defined on every eligible-tail and frontier cut path. This occurrence lift is absent. The four downstream maps are consequently untestable on the actual carrier, although their abstract kernel/fiber and commuting-square criteria are proved.

On the arithmetic side, the exact identities

$$t = aD(z), \quad t + 2 = sV(z), \quad \mu(D(z))\mu(V(z)) = G_a(t)G_s(t + 2)$$

place a bounded periodic core in the quantitative affine corridor. This is real arithmetic progress, classified as L1_{actual-core}. It is not positive L2: physical weights, generic phases, all-prefix selection and the complete four-sign return remain missing, and its fixed- X -power exponent is zero.

A canonical cross-platform source manifest, a typed proof DAG, a minimal-missing-antichain algorithm, a role-based H9 firewall and separate arithmetic, physical and full-synthesis endpoint ledgers freeze the conclusion. The first missing node is H1.frontier_occurrence_lift; no 1/400 endpoint, hard-packet estimate or prime-pair theorem is obtained.

Keywords: fixed shift; occurrence lift; frontier totalization; Möbius correlation; almost all scales; endpoint certificate.

Claim boundary. The source hashes detect drift but do not prove the hashed theorems. The identity $P_{h_0}^{\text{cut}} = I$ is not a downstream prescribed-shift theorem. An almost-scale logarithmic saving on an actual Möbius core is not an all-prefix physical saving and has zero exponent on the fixed- X -power ledger. Nothing below proves $B_{h_0, \delta} = o(X)$, a prime-pair lower bound, or the twin-prime conjecture.

1 The two inherited openings

TPC-141 retained the exact cut identity

$$B_{h_0, \delta}(X) = S_{\text{soft}}(X) + S_{\text{elig}}(X) + S_{\text{front}}(X), \quad S_{\text{soft}}(X) = o(X),$$

and TPC-142 returned NOT-TESTABLE because the frontier was not totalized [2, 3]. The next batch separates the missing frontier object from the known arithmetic limitations.

Let $\mathcal{C}_X^{\text{ns}}$ contain *all* nonsoft cut paths:

$$\mathcal{C}_X^{\text{ns}} = \mathcal{C}_X^{\text{ETO}} \sqcup \mathcal{C}_X^{\text{FUM}},$$

where ETO denotes `ELIGIBLE_TAIL_OPEN` and FUM denotes `FRONTIER_UNMAPPED`. The frozen finite sample contains 2,988 FUM paths and no ETO path. This sample fact cannot remove ETO from the asymptotic domain.

The arithmetic branch starts from the determinant-two affine pair

$$D(z) = d + sz, \quad V(z) = u + az, \quad su - ad = 2.$$

TPC-147–150 test whether the quantitative affine theorem of Tao–Teräväinen can control a literal core of this pair [1, 8–11].

2 Canonical source locking

Raw byte hashes are unsuitable for a repository used on both Windows and Linux: generated JSON may have CRLF or LF line endings while representing the same canonical object. We therefore use the rule `CANONICAL_UTF8_LF_V2`.

Definition 2.1 (Canonical source bundle). Text is decoded strictly as UTF-8 and CRLF or CR is normalized to LF. JSON is parsed and rendered with sorted keys and one final LF. Relative paths use POSIX separators and are sorted. The bundle hash is the SHA-256 digest of the canonical map of relative paths to canonical content digests. The bundle does not contain its own hash.

The manifest anchors the canonical TPC-141 manifest through the hash embedded in the TPC-142 snapshot, and then source-locks TPC-143–150. Tuple, path and occurrence identities remain exact typed objects. No collision-resistance assumption is used as a mathematical injectivity theorem.

Theorem 2.2 (Non-escalating typed merge). *Suppose every proof import matches one source export in status, evidence level, direction, role, scope, carrier, normalization and coverage; the proof DAG is acyclic; and every shadow-only export has no promotion edge. Then the merged conclusion is a consequence of the imported conclusions in their recorded scopes. Canonical source hashes contribute integrity only and cannot strengthen that conclusion.*

Proof. Induct along the canonical topological order. Exact type matching justifies each invoked edge, while the absence of a promotion edge prevents a scope-mismatched result from entering the selected root. The hash is not a logical edge and therefore supplies no theorem premise. $\square$

3 The occurrence lift and four-map contract

TPC-143 distinguishes the cut selector from every downstream selector and proves

$$\text{H1.cut_shift_selector} = \text{PROVED}_{L1}, \quad P_{h_0}^{\text{cut}} = I$$

[4]. It also proves a field-descent criterion: a downstream label descends through the current forgetting projection if and only if it is constant on every projection fiber. Legal completions with the same cut record and different determinant, zero-mode, grouping or shift-lineage labels show that the four labels do not descend within the maximal formal completion class constrained only by the currently archived fields and identities. This is a current-schema route obstruction, not a theorem that the unknown actual-support carrier admits two such completions or that an augmented construction is impossible.

Definition 3.1 (Occurrence lift). An occurrence lift is a sparse row-separated matrix

$$L_X : \mathbb{C}^{\mathcal{C}_X^{\text{ns}}} \longrightarrow \mathbb{C}^{\mathcal{O}_X}.$$

One cut column may have several occurrence rows. Each nonzero edge records exact source, target, occurrence, parent, stage and local-edge identifiers, its multiplier AST, canonical-parent lineage, native lineage, source and target shift tags, normalization, support status, mask and boundary data. It is conservative when

$$\mathbf{1}^T L_X = \mathbf{1}^T.$$

The one-to-many nature is essential: a single canonical parent may have several computational or physical occurrences. Conversely, every target occurrence must have a unique recorded canonical-parent lineage. Formal-envelope, canonical-parent and actual-active support statuses are different namespaces.

TPC-144 gives abstract determinant and zero-mode quotient interfaces [5]. For surjective completed maps Q_D, Q_Z , a unique isomorphism J with $JQ_D = Q_Z$ exists exactly when

$$\ker Q_D = \ker Q_Z.$$

Literal fiber relabeling additionally requires equality of the occurrence equivalence relations and exact edge multipliers. Equality of one aggregated scalar is insufficient.

TPC-145 gives the occurrence-resolved prescribed-shift square [6]. On an aggregated matrix G ,

$$P_{\text{phys}}G = GP_{\text{src}}$$

may vanish by cancellation between two leaking edges. On the row-separated lift, commutation is equivalent to edgewise preservation of the predicate “shift equals h_0 .” Thus $P_{h_0}^{\text{cut}} = I$ cannot be substituted for a downstream selector.

TPC-146 combines these tests into the defect vector

$$\mathfrak{D}_X = (D_L, D_{Q_D}, D_{Q_Z}, D_G, D_P, D_{DZ}, D_{GP}, D_{\text{cover}}, D_{\text{rec}}).$$

It preserves two distinct sufficient routes:

$$\begin{aligned} \text{H1Complete}(X) \iff \text{MapTotal}_{\text{ns}}(X) \\ \vee (\text{ScalarSoft}_{\text{FUM}}(X) \wedge \text{ETOTotal}(X)). \end{aligned}$$

The second disjunct requires both $S_{\text{front}}(X) = o(X)$ for the complete original frontier sum and a theorem-backed disposition of every eligible-tail-open path: total downstream maps or a complete ETO soft theorem. The empty ETO class in one finite sample cannot discharge this second clause. Neither disjunct is presently proved [7].

Proposition 3.2 (Current structural pointer). *In the selected occurrence-augmented map route, the minimal missing antichain is the singleton*

$$\boxed{\{\text{H1.frontier_occurrence_lift}\}}.$$

All actual $Q_D, Q_Z, G, P_{h_0}^{\text{down}}$ totality and intertwining nodes are its descendants.

4 A real actual-core arithmetic corridor

Define the quotient lift G_c on multiples of c by

$$G_c(cm) = \mu(m).$$

For $t = aD(z) = ad + asz$, the determinant relation gives

$$t + 2 = sV(z),$$

and hence the exact identity

$$\mu(D(z))\mu(V(z)) = G_a(t)G_s(t+2). \tag{1}$$

TPC-147 supplies periodic residue reassembly; TPC-148 verifies the quotient lift and the relevant nonpretentiousness; and TPC-149 applies the quantitative affine theorem to the small-polylogarithmic periodic core [1, 8–10].

Proposition 4.1 (Exact evidence classification). *The result imported from TPC-149 is*

$$\text{A149.actual_mobius_periodic_corridor} = \text{PROVED}_{\text{L1}_{\text{actual-core}}}.$$

It is a theorem about actual Möbius factors in its stated periodic core and almost-scale scope. It is not a theorem on the full physical H3 carrier and its fixed- X -power exponent is 0.

The distinction avoids two opposite mistakes. Calling (1) a synthetic model would discard a real arithmetic advance. Calling its logarithmic almost-scale saving positive L2 would silently add physical weights, generic phases, deterministic prefixes and a four-sign return that have not been proved. TPC-150 records precisely these return obligations and a scoped obstruction to atomic prefix selection [11].

5 Route cells and the first missing antichain

Each tested negative result stops only its exact route cell. The eligible-only compiler, direct shift-one reparameterization, current-schema-only downstream derivation, schema-only Q_D/Q_Z derivation, aggregate-cancellation shift route and atomic-prefix route are stopped in their declared scopes. None stops the selected occurrence-lift/quotient-Möbius route.

The route list is a declared collection of tested cells, not a theorem that every conceivable architecture has been enumerated. Therefore, even a future stop of every listed cell would not establish architecture infeasibility without an independent route-universe completeness theorem.

Definition 5.1 (Minimal missing antichain). Let $\mathcal{A}$ be the ancestor closure of the selected root, including the root. A node is missing if it is NOT-TESTABLE or fails scope matching. The minimal missing set contains the missing nodes having no missing ancestor in $\mathcal{A}$. For deterministic reporting only, its first representative is chosen by canonical topological rank and then node identifier.

This definition prevents a missing descendant from pre-empting its missing prerequisite and does not pretend that incomparable missing nodes have a mathematical linear order. Applied to the frozen DAG it gives [proposition 3.2](#).

6 H9 and the split endpoint

The physical occurrence registry is structural. Its ancestor closure may contain H1, H6, H7, H8 and independent physical-synthesis cost theorems, but no node whose explicit role is arithmetic shadow, arithmetic target or arithmetic negative. A second check excludes the H2–H5 gate prefixes. Thus relabeling an arithmetic node cannot evade the H9 firewall.

The old phrase “the endpoint” hides two different tasks. We record

$$\begin{aligned} \text{arithmetic target:} \quad & \sigma_{\text{raw}} \geq 1/400, \\ \text{physical loss target:} \quad & \Lambda_{\text{phys}} < 1/400. \end{aligned}$$

The first may remain open at the arithmetic frontier; the second must be certified from a complete H9 registry independently of H2–H5. Full synthesis additionally requires strictly positive net slack.

Proposition 6.1 (Directional endpoint calculus). *For a complete physical route cell, a theorem-backed upper certificate $U < 1/400$ passes the physical-loss endpoint. A theorem-backed lower or exact dual certificate $L \geq 1/400$ stops that cell. An upper certificate $U \geq 1/400$ without such a lower certificate proves neither pass nor stop. Unknown costs are not zero, energy exponents are halved exactly once, joint tokens replace exact disjoint covers, and determinant reserve is not a physical token.*

The actual-core logarithmic saving is stored separately with fixed- X -power exponent 0. It cannot be subtracted from Λ_{phys} or used to pass either $1/400$ contract.

7 Source-locked integration theorem

Theorem 7.1 (TPC-151 integration status). *For the canonical source-locked TPC-143–150 batch:*

- (i) *the structural branch establishes the exact occurrence-lift and four-map completion contract, but the actual lift is NOT-TESTABLE;*

- (ii) the arithmetic branch establishes a fixed-two Möbius periodic core in its small-polylogarithmic almost-scale scope, classified as $L1_{\text{actual-core}}$;
- (iii) the physical and full-synthesis endpoint ledgers are incomplete; and
- (iv) the first missing node is `H1.frontier_occurrence_lift`.

Consequently the integration is assembled with gaps and contains no positive L2 theorem.

Proof. The typed merge theorem preserves every source scope. The map route cannot pass because L_X and its descendant defects are unknown; the scalar route cannot pass because neither a complete $S_{\text{front}} = o(X)$ theorem nor a theorem-backed ETO disposition is imported. The TPC-149 export has scope `scope.actual_fixed_two_mobius_periodic_core_almost_scale`, has no promotion edge and records zero fixed- X -power exponent. The occurrence registry is incomplete, so the physical-loss endpoint is also incomplete. The active-closure scan then gives the stated minimal missing singleton. $\square$

Corollary 7.2 (Precise progress). *The batch contains both structural L1 progress and genuine actual-core arithmetic L1 progress. Neither is fixed- h_0 physical positive-L2 progress.*

8 Deterministic audit

The standard-library audit checks canonical LF hashing, the TPC-141/TPC-142 anchor chain, exact imports, the proof DAG, the active ancestor closure, the minimal missing antichain, role- and gate-based H9 independence, scoped route stops and rational endpoint directions. Mutation tests reject:

1. line-ending-dependent hashes and typed import mismatches;
2. current-schema formal nonidentifiability promoted to an impossibility statement on the actual carrier;
3. the $L2_{\text{actual-positive}}$ target label reported as an achieved result;
4. an actual-core pseudo-L2 promotion;
5. an arithmetic dependency hidden below H8;
6. an empty ETO sample used as an asymptotic totality certificate;
7. a frontier-totalization artifact compressed to a scalar theorem without its complete ETO disposition;
8. an endpoint artifact that drops its actual-carrier scope or strict $1/400$ direction;
9. $P_{h_0}^{\text{cut}}$ used as the downstream selector;
10. an unproved route-universe completeness flag; and
11. upper-bound failure, endpoint equality and unknown-cost direction errors.

Established. The source-lock, occurrence-lift contract, four-map audit, exact actual-core evidence classification, H9 firewall and split endpoint calculus are L0/L1 results.

Not established. No occurrence lift, complete four-map route, scalar frontier theorem or complete ETO disposition, all-prefix physical return, positive fixed- X -power saving, strict physical endpoint, $B_{h_0, \delta} = o(X)$, prime-pair lower bound or twin-prime theorem is proved.

9 Conclusion

The first unsupported carrier is no longer represented by a generic “missing maps” label. It is the concrete conservative matrix L_X , with a complete field and defect contract. In parallel, the quotient lift reaches a genuine fixed-two Möbius periodic core without pretending that an almost-scale logarithmic theorem is the full physical estimate. These are two forward moves at different levels. Their source-locked combination still stops before the physical carrier, so the next route decision must remain NOT-TESTABLE.

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